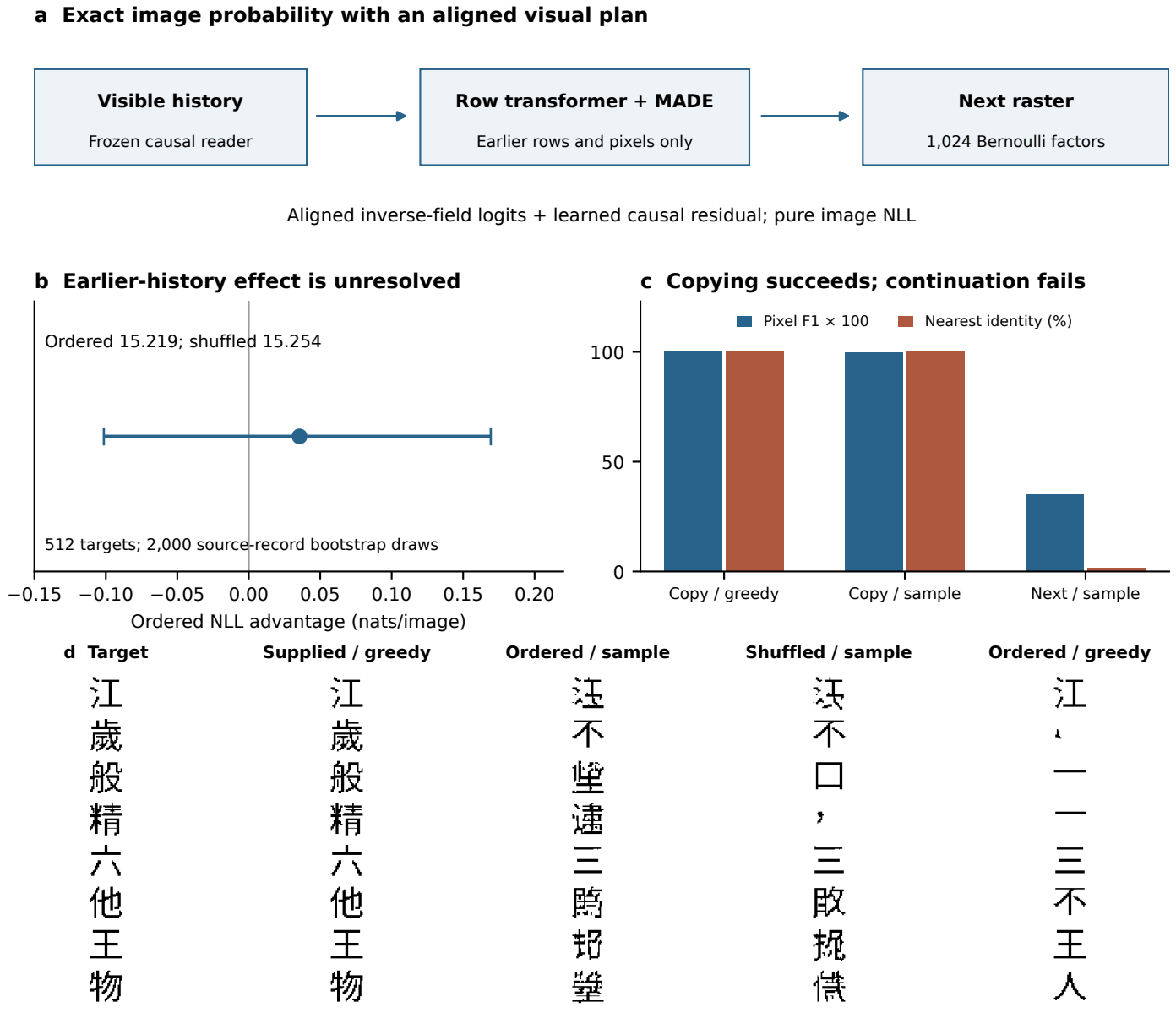
a Exact image probability with an aligned visual plan
Visible history
Frozen causal reader
Row transformer + MADE
Earlier rows and pixels only
Next raster
1,024 Bernoulli factors
Aligned inverse-field logits + learned causal residual; pure image NLL
b Earlier-history effect is unresolved
Ordered 15.219; shuffled 15.254
512 targets; 2,000 source-record bootstrap draws
−0.15 −0.10 −0.05 0.00 0.05 0.10 0.15 0.20
Ordered NLL advantage (nats/image)
c Copying succeeds; continuation fails
Pixel F1 × 100
Nearest identity (%)
100
50
0
Copy / greedy
Copy / sample
Next / sample
d Target
Supplied / greedy
Ordered / sample
Shuffled / sample
Ordered / greedy